Set 24: Product Inventory Management

AIM

To analyze product inventory using R programming by creating a bar chart and a stacked bar chart to visualize available quantities, and a scatter plot to examine the relationship between product price and quantity available.

ALGORITHM

1. Create a dataset containing Product ID, Product Name, and Quantity Available.
2. Load the ggplot2 library.
3. Create a bar chart to visualize the available quantity of each product.
4. Create a stacked bar chart to display product quantities as segments of total inventory.
5. Create a scatter plot to explore the relationship between product price and quantity available.
6. Interpret the inventory distribution and the relationship between price and quantity.

PROGRAM

```
library(ggplot2)


inventory <- data.frame(
  Product_ID = 1:5,
  Product_Name = c("Product A", "Product B", "Product C",
                   "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220),
  Price = c(100, 150, 80, 120, 90)
)


ggplot(inventory, aes(Product_Name, Quantity, fill = Product_Name)) +
  geom_col() +
  labs(title = "Available Product Inventory",
       x = "Product Name", y = "Quantity Available")


ggplot(inventory, aes(x = "Total Inventory",
                      y = Quantity, fill = Product_Name)) +
```

```
  geom_col() +
  labs(title = "Product-wise Share of Total Inventory",
       x = "Inventory", y = "Quantity Available")

ggplot(inventory, aes(Price, Quantity, color = Product_Name)) +
  geom_point(size = 3) +
  labs(title = "Product Price vs Available Quantity",
       x = "Product Price", y = "Quantity Available")

print(inventory)
```

Note: Product prices are illustrative because the given dataset does not include prices. The stacked bar chart shows each product's contribution to total inventory.

OUTPUT:



RESULT

The bar chart compares the available quantities of all five products. Product C has the highest inventory with 300 units, while Product B has the lowest with 175 units. The stacked bar chart shows each product's share of the total inventory, and the scatter plot explores the relationship between product prices and available quantities.